

Kingdom of Saudi Arabia
Qassim University
College of Computer



المملكة العربية السعودية
جامعة القصيم
كلية الحاسب

CS432

AI PROJECT



Ragheed Almasry – 412117664

Anas Alarbeed – 411116054

Suliman Alfawzan – 412111550

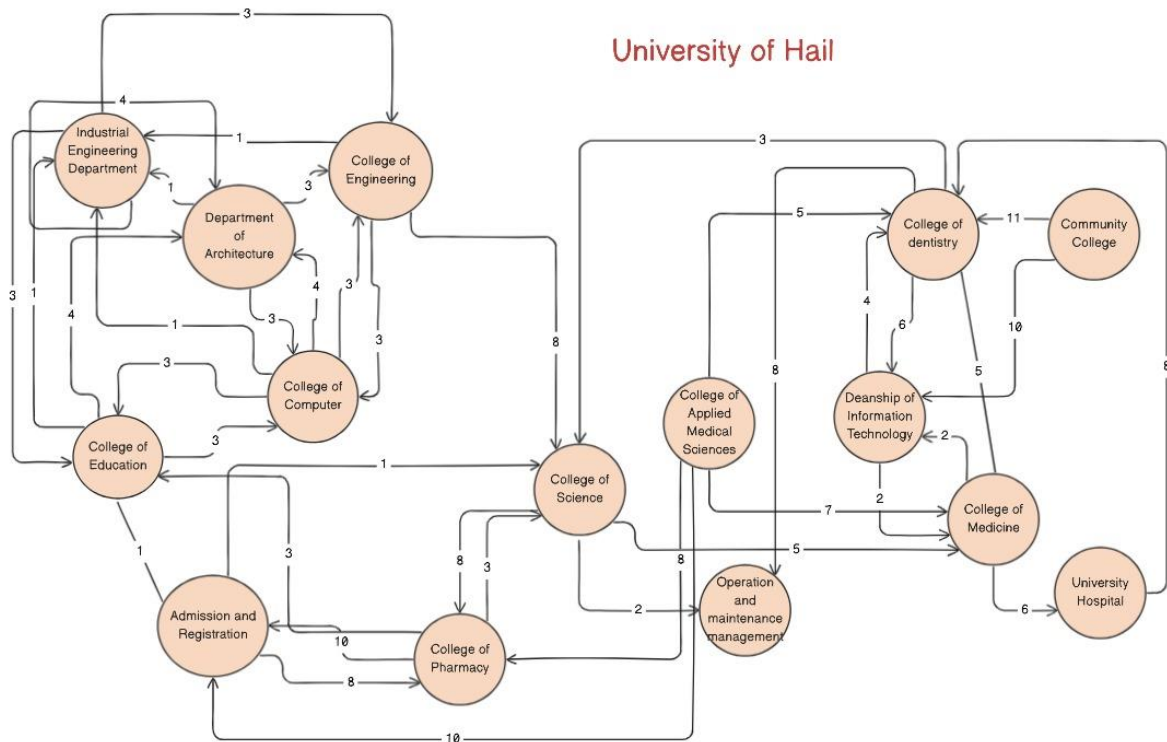
Supervisor: Dr. Abdalla Musa

CS432

Intelligent Route Searching for a Postal Robot of Hail University

1. The Graph represents Hail University

- The campus consists of 15 buildings.



2. The three search strategies used are

- Uninformed
 - Depth First Search (DFS)
 - Breadth First Search (BFS)
- Informed
 - Greedy Heuristic Search

3. The Language used is Python

4. The Program results are:

- From College of Medicine to College of Science
 - Output:

```
DFS:
-----Navigation from College of Medicine to College of Science-----

Visited nodes:
College of Medicine, College of dentistry, College of Science,

Path: ['College of Medicine', 'College of dentistry', 'College of Science']
Cost of the path: 8
Time taken for DFS: 0.111200 milliseconds

BFS:
-----Navigation from College of Medicine to College of Science-----

Visited nodes:
College of Medicine, College of dentistry, Deanship of Information Technology, College of Science,

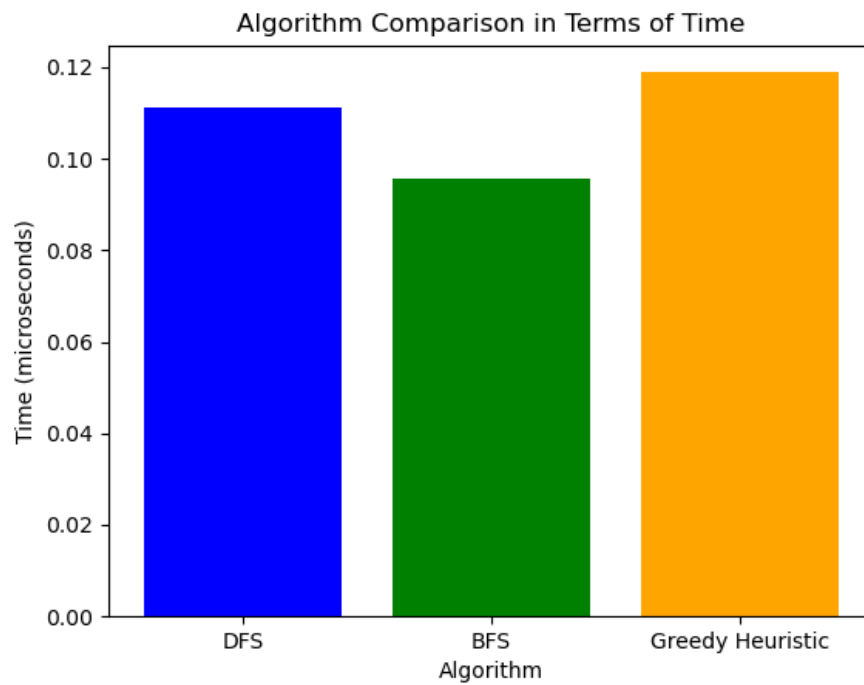
Path: ['College of Medicine', 'College of dentistry', 'College of Science']
Cost of the path: 8
Time taken for BFS: 0.095600 milliseconds

GREEDY:
-----Navigation from College of Medicine to College of Science-----

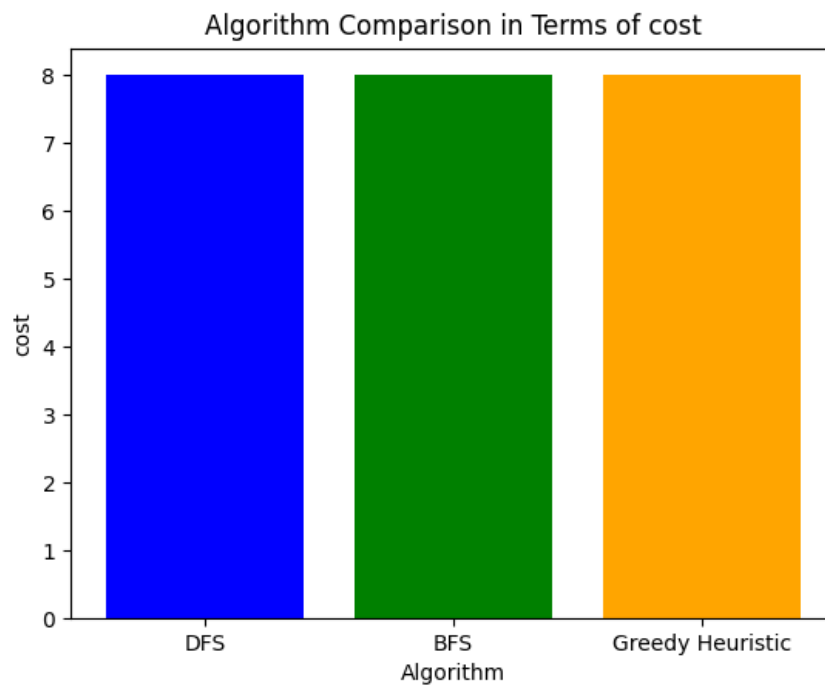
Visited nodes:
College of Medicine, College of dentistry, College of Science,

Path: ['College of Medicine', 'College of dentistry', 'College of Science']
Cost of the path: 8
Time taken for Greedy Heuristic: 0.118900 milliseconds
```

- **Comparison in term of Time:**



- **Comparison in term of Cost:**



- From College of Computer to College of Pharmacy

- Output:

```
DFS:
-----Navigation from College of Computer to College of Pharmacy -----

Visited nodes:
College of Computer, College of Education, College of Engineering, College of Science, College of Medicine,
College of dentistry, Deanship of Information Technology, Operation and maintenance management, Deanship of Information Technology, College of Pharmacy ,

Path: ['College of Computer', 'College of Education', 'College of Engineering', 'College of Science', 'College of Pharmacy ']
Cost of the path: 22
Time taken for DFS: 0.249500 milliseconds

BFS:
-----Navigation from College of Computer to College of Pharmacy -----

Visited nodes:
College of Computer, College of Education, College of Engineering, Department of Architecture, Industrial Engineering Department,
College of Engineering, Department of Architecture, Industrial Engineering Department, College of Science, Industrial Engineering Department,
Industrial Engineering Department, College of Science, College of Medicine, College of Pharmacy ,

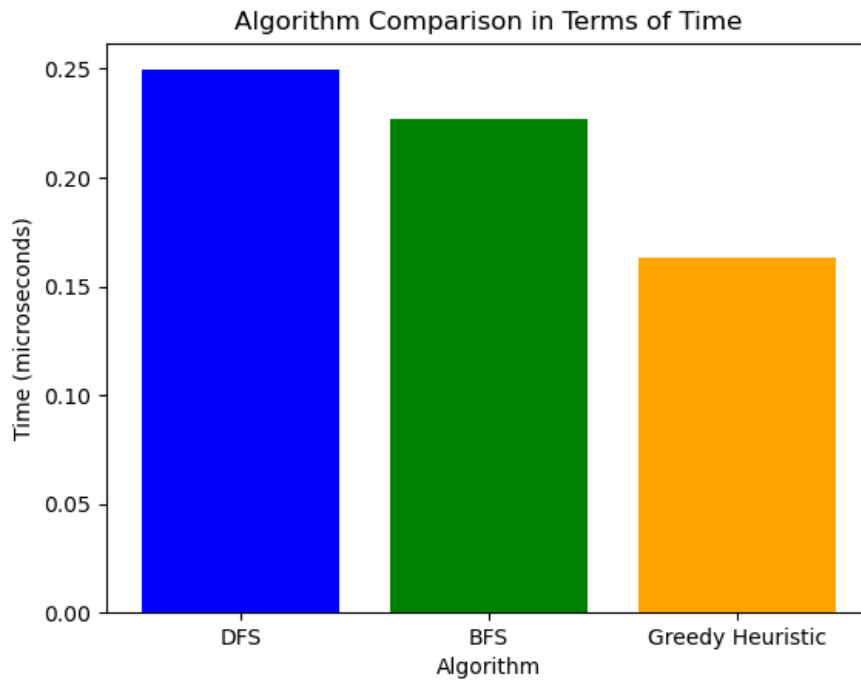
Path: ['College of Computer', 'College of Engineering', 'College of Science', 'College of Pharmacy ']
Cost of the path: 19
Time taken for BFS: 0.226800 milliseconds

GREEDY:
-----Navigation from College of Computer to College of Pharmacy -----

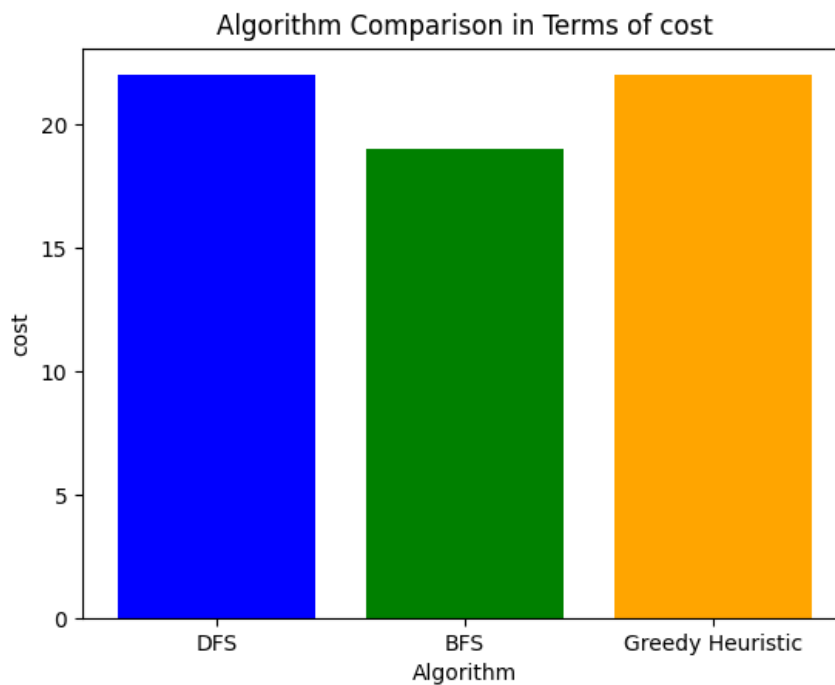
Visited nodes:
College of Computer, College of Education, Department of Architecture, Department of Architecture, College of Engineering,
College of Science, College of Pharmacy ,

Path: ['College of Computer', 'College of Education', 'College of Engineering', 'College of Science', 'College of Pharmacy ']
Cost of the path: 22
Time taken for Greedy Heuristic: 0.163300 milliseconds
```

- **Comparison in term of Time:**



- **Comparison in term of Cost:**



- From College of Engineering to Department of Architecture

- Output:

```
DFS:
-----Navigation from College of Engineering to Department of Architecture-----

Visited nodes:
College of Engineering, College of Computer, College of Education, Department of Architecture,

Path: ['College of Engineering', 'College of Computer', 'College of Education', 'Department of Architecture']
Cost of the path: 10
Time taken for DFS: 0.273000 milliseconds

BFS:
-----Navigation from College of Engineering to Department of Architecture-----

Visited nodes:
College of Engineering, College of Computer, College of Science, Industrial Engineering Department, College of Education,
Department of Architecture,

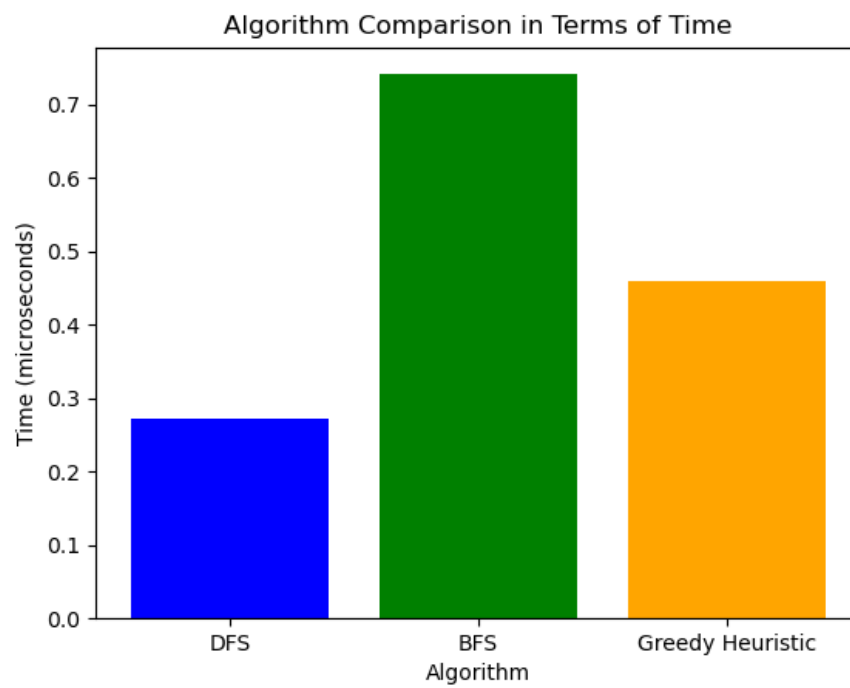
Path: ['College of Engineering', 'College of Computer', 'Department of Architecture']
Cost of the path: 7
Time taken for BFS: 0.741100 milliseconds

GREEDY:
-----Navigation from College of Engineering to Department of Architecture-----

Visited nodes:
College of Engineering, Industrial Engineering Department, Department of Architecture,

Path: ['College of Engineering', 'Industrial Engineering Department', 'Department of Architecture']
Cost of the path: 5
Time taken for Greedy Heuristic: 0.459400 milliseconds
```

- **Comparison in term of Time:**



- **Comparison in term of Cost:**

